

Design and Implementation of Mobile-based Application for Appointments Systems for Vaccinations in Medical Centers

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Abstract— The current patient registration and operating procedure in medical centers are time-consuming and somehow tiresome for patients and their families. Normally, patients have to visit the medical center or hospital to register, they have to bring their medical records, and wait to be called by the medical staff. The situation becomes worse when the patients are kids because they cannot wait for long time and their parents do not have enough time to wait for long time. In view of these problems, several appointment applications are proposed to enhance the registration systems and reduce the waiting time. However, these efforts are still face some drawbacks and limitations. In this paper, we have designed an Android-based mobile application to create appointments with doctors in medical clinics and store the patient's health records. This mobile application is integrated with an interactive website and a comprehensive database to enable the doctors to handle vaccinations appointment, maintain and retrieve patients' medical records. The managers also can efficiently create, manage and admin the clinic's work progress. This mobile application will enhance appointment scheduling and improve the quality of healthcare. In addition, the application can reduce the time and effort consumed to perform the vaccinations appointment process and taking the vaccinations on time.

Keywords—: Android Application, Website Design, Healthcare Centers, Electronic Record, Vaccinations Appointment.

I. INTRODUCTION

We are aware of the difficulties and issues faced when visiting a hospital or a medical center to book an appointment for check-up or to take vaccinations. During this visit, you have to register and wait on long queues at the reception desk that consumes a lot of time. It also becomes difficult for senior citizens to personally visit a hospital or clinic to book an appointment for themselves. The problem of registration becomes worse when the registration related to the childrens to take vaccinations or visit the doctor's because they cannot be silent for long time and they cannot wait for long time without milk.

Nowadays, the parents face problems to follow up their children growth; they need to remember all the details about

all vaccines up to six years. Some times they forget the dates, which may cause serious diseases in the future.

Child health care is considered as one of the foremost priorities in the world. However, it is an undeniable truth that millions of children's around the world are affected each year by disease. Infectious diseases have been considered as a major cause of morbidity and mortality in children under 5 years. According to a recent report announced by the World Health Organization (WHO), 83% of the global deaths of children under 5 years are caused by infectious diseases and malnutrition [1].

Vaccines can help to protect infants and children's from serious diseases. Getting childhood vaccines means your child can develop immunity against diseases. Generally young children are at increased risk for infection diseases because their immune systems have not yet built up the necessary defenses to fight serious infections and diseases. As a result, its important manage the vaccinations for childrens and monitor the health history for each child.

Health information record of a new-born is important for quality healthcare. Ideally, it should comprise digital record of growth, periodic health examinations and vaccinations of a child [2]. In this paper, we provide a solution by combining database technologies with mobile computing to imrove the quality of service provided for patients and medical staff in medical clinics. For this purpose, we have designed an Android-based mobile application and interactive website to enable parents to create appointments for taking vaccinations for their childrens. This application enables the parents to book appointment for vaccinations and view upcoming vaccinations. The parents received a notification alert to remember them of the upcoming appointment. In addition, they can view more details about infectious diseases, including corresponding symptoms, causes, incubation period, preventive measures, nutritional guidelines such as breastfeeding and information about diagnosing infectious diseases based on symptoms. The developed website can help the medical staff to manage the vaccinations schedule, editing the prescription, receiving medical indications based on the entered data. This should improve the medical services provided for patients by reducing the waiting time in queues and having doctor consultation.

We organize the rest of this paper as follows. In section II, we discuss some related work on mobile health care systems. The problem statement and detailed description of the developed system is give in Section III and Section IV respectively. Finally, we conclude our work and discuss some future works in Section V.

II. RELATED WORKS

In the past few years, there is increasing accessibility to mobile devices due to the reduced pricing for smartphones and tablets and the unprecedented processing power and huge storage capacity of the mobile devices. This development increases the number of applications developed to help the community including the health sector, mobile banking, mobile shopping, travel management and many other fields. Recently, many medical appointment applications for smart phones have been developed and widely used by health sector and patients [3]. These applications is very useful because it leads to better communication between patients and doctors and enhance the overall treatment quality; here we reviewed the recent research papers and mobile applications.

In [4] a mobile based doctor patient interaction system was proposed to allow patients to register by login to the system and book an appointment with a doctor, and can also make complaints on the system. Doctors can also login into the platform, view complaints and as well pot solutions to complaints, also view the clinic register.

An Android Application for Appointment booking System prposed by [5]. The system proposes two main modules that include the Doctor and the patient. The patient can book an appointment with a doctor after reviewing doctor specialization, the doctors' profile and view the doctors schedule so as to know when to book appointment with the particular doctor. The doctor can either accept or reject the appointment.

The authors in [6] proposed a real-time appointment scheduling to enable the patient to schudle appointment on a fixed time and date and allocate a doctor available on that particular time and date. The system can also handle the rescheduling of patients with doctors. In addition, the system integrates a live consultation with a doctor available online. This system adopt object-oriented analysis and design approach and implement an android mobile application using JAVA as the programming language and use PHP as frontend while MySQL used for the backend.

In [7], the authors developed an Android mobile application to overcome the difficulties in traditional booking system by introducing an on-the-go appointment booking system which helps patients schedule an appointment with their preferred doctor. The system cosists of two portals, doctor and patient. The patient enter his information by himself on the first time of use and stored on the database for future login. They can view their profile, appointments doctor's details and they can schedule the appointment in the clinic or hospital they are currently practicing. This system also enables the patient to select the available time slot for booking an appointment.

III. PROBLEM STATEMENT

Recently, parents are very busy due to their work, housework and other activities that make them busy all the time and forget important issues such as their new-born health

care. The new-born needs to vaccinate and check that the growth is normal. This treatment and monitoring are done through visiting the medical clinic. To visit the clinic, you have to come on time and the have to wait for long times and registration and appointments to take these vaccines. Sometimes the parents forget the vaccinate dates, which may cause serious diseases in the future.

A study was done on the most important problems faced by mothers in caring for children, it was noted that lately there are many problems facing children, and the most important of these problems is malnutrition or unhealthy growth of kids [2].

Also, the problem is the delay in discovering any defect in the child's hearing, speech, or vision, especially in the early months of the child's age, or even the delay in discovering a child's talent.

All of these problems are important reasons for the lack of parental knowledge, one of the serious problem that currently facing mothers is forgetting the date of the vaccination, or the date of measuring the length and weight of the child, especially if she is a working mother, and because there is no integrated child care application. Based on the fore mentioned problems, an application is developed to reduce these problems that can affect the new-born growth.

IV. PROPOSED APPROACH

The proposed system, which we called "ThanksMamy" consists of a mobile application, website and central database. In the proposed system, the doctors, parents, medical and adminstrative staff are brought to one place which is handled by system admin. The Android application is mainly used by the parents and the website is used by all other parties and for parents who don't has a mobile or prefer to use the website rather than the mobile application.

A. System Architecture

The system architecture consists of three main layers Presentation Layer, Service Layer and Resource Layer As shown in fig. 1.

The Presentation layer shows the user interface for logging to the system by assigned access rights for each user. The website, which is used by doctors, nurses, manager, and parents in special cases, and the mobile application which used by parents. Parents can use the Android application to manage their appointments and view important information about their children. This mobile application synchronized with the website using the web service and cenralized database. Through the website, the doctor can view his appointment list, schedule the appointments, consult the medical record, write a prescription, and view the medical history of the registered childrens. The managers can admin the work process of the medical clinics. Also, the parents can take advantage of the health system if they don't have a smart phone to check vaccination details, schedule an appointment and all services provided through the mobile application. The system admin can use the website to admin the whole system including the website, database and perform maintinance and updates of the system.

The *Service Layer* is essential for our system and it performs services to the system units. It contacts with the Resource Layer and assemble information from the requested database. The third layer is the Resource Layer, which

contains the system databases and respond to the requests from the database users.

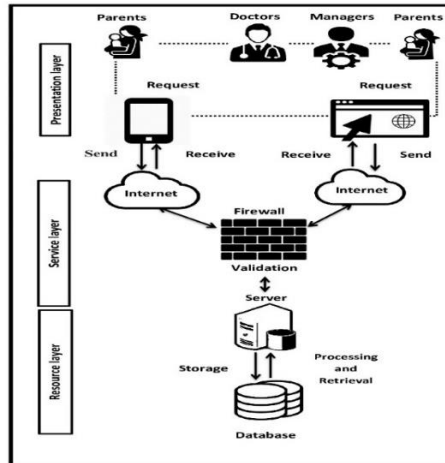


Fig. 1: Over view System Architecture

B. Mobile Application Design

The patient parents need to visit the medical clinic for one time registration since the medical history record include medical information that parents may not be able to fill in the registration form such as blood type, RH factor, height at birth, weight at birth, apgar score, meconium passage, episiotomy, perineal tear, blood transfusion and congenital malformation. After registration, users can login to their respective profiles using unique user ID and password. The proposed application facilitates to change the password after first login.

To implement this mobile application, several programming languages were used including Android Studio, Java & Kotlin, XAMPP Apache HTTP server, Maria DB, MySQL, JSON, Firebase and an interpreter for applications written in PHP. Figure 2. Shows the architecture of the mobile system.

Figure 2. Shows the architecture of the mobile system. The user can use the android application to retrieve data from the MySQL database by sending Http Request, then the application process the Request with server side using PHP script files. PHP script files are located in the middle between the android application and MySQL database and perform communication using POST/GET commands. JSON object is used to handle incoming data from the MySQL database and encode/decode the result in JSON representation format. The results for PHP scripts is return back to the Android application.

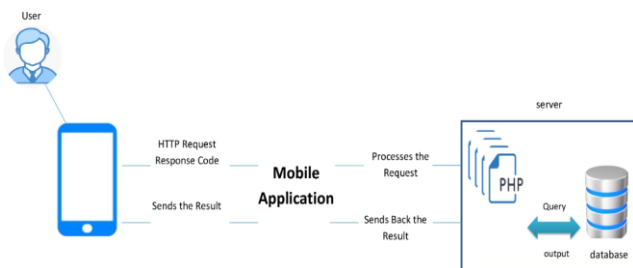
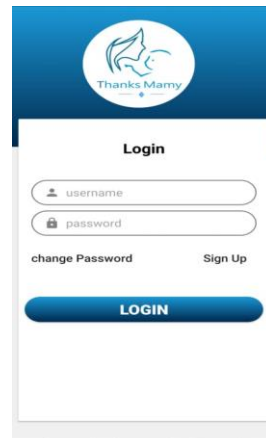


Fig. 2: Overview of Mobile Application Architecture

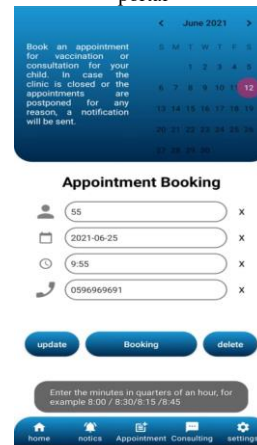
The mobile application has many activities including login/logout, appointment booking, medical health record, medical consultation and many other interesting activities as shown in Fig. (3).



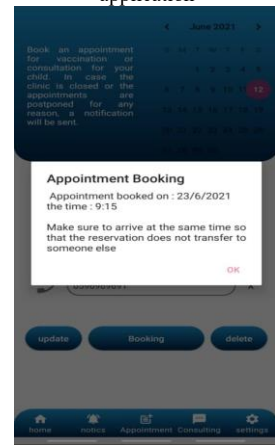
(a) login/logout screen for patients portal



(b) main screen of mobile application



(c) Appointment Booking



(c) Booking confirmation

Fig.3. Mobile Application

C. Website Implementation

To develop the website, the following tools is used including Visual Studio, Apache HTTP server, Maria DB, MySQL, PHP, Ajax, HTML, CSS, JavaScript and Bootstrap. The website is built enable the administrators to manage the system. The management staff, parents, doctors and nurses can use login page authenticate their access to the database through the website for different purposes as shown in Fig. 4

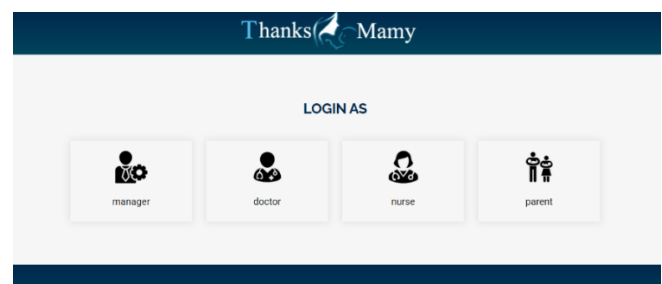


Fig. 4: website login page

- Doctor Module

The doctors can login using their user ID and passwords that provided explicitly by the medical authorities. After login they will redirected to their respective profiles. The doctors can use the website to perform different services including view the information of the registered children in the clinic and view their medical history. They can view the booked

appointments in the system and approve all the requested appointments sent by the parents. Also, they can add prescribed medications and diagnosis about the vaccination and store them in the medical history. In addition, they can post notifications and announcement for parents and other services. Nurses also can use the website to perform different services that they can perform during their work like child registration as shown in Fig. 5.

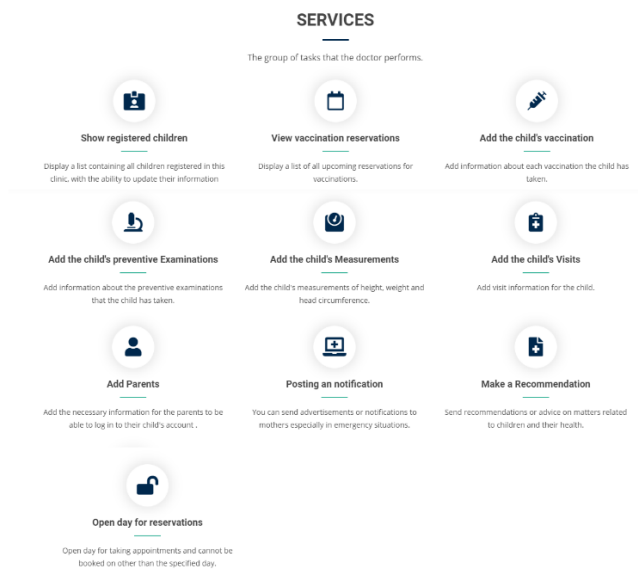


Fig. 5: Doctors services on the website

- Manager Module

The manager of the health clinic can use the website to perform different tasks including add new doctors/nurse, add announcements and information for parents, view the appointments for each clinic and view the profiles of the registered children as shown in Fig. 6.

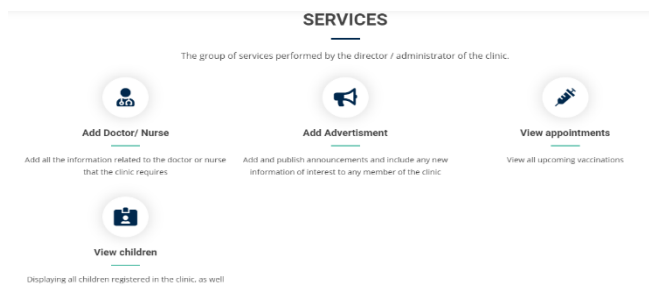


Fig. 6: Medical Authorities services on the website

-Parents Module

Parents can also login to the website to use similar features as those used by the mobile application using the same user-ID and password used for the mobile application. Parents can perform all the activities available on the mobile application by entering to their profile on the website. This option is added to the parents since some of them may not have a smart phone or face difficulties to use the mobile application to perform several services as shown in Fig. 10 and Fig. 7.

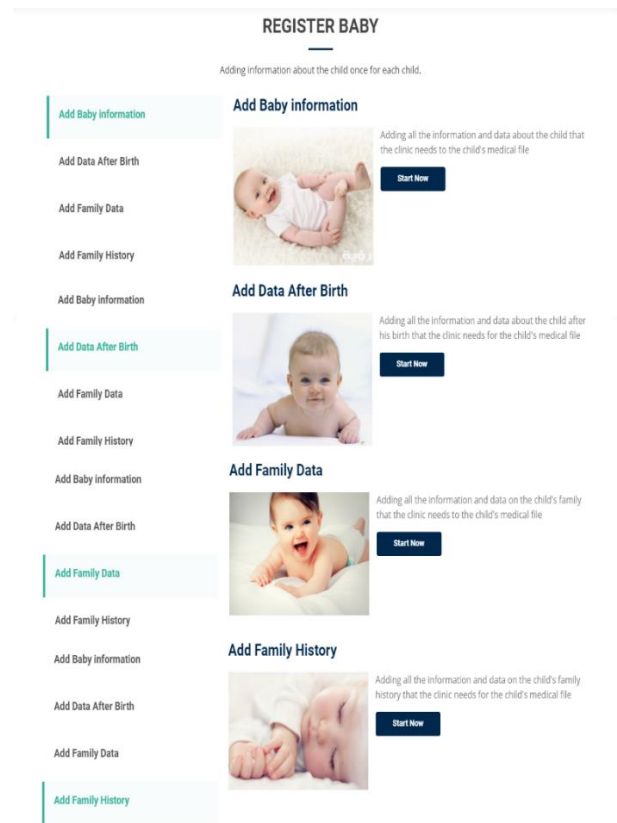


Fig. 7: Parent service on the website

D. Database system

A comprehensive database developed to help the system users to access and store the appointments information, medical profile, medication history and retrieval of much more information. The database is synchronized with the both the website and the Android application using MySQL in order to enable the users to store, access and retrieve requested data any time any where.

The developed android application can communication with a remote MySQL database to authenticate login information and also to retrieve/ update medical information of the patients in the database. The android application

Use HTTP protocol to make a connection with a PHP script located in the server. PHP scripts are associated with specific tasks between the android application and MySQL database using POST/GET. JSON object on the other hand is used to handle incoming data from the MySQL database and encode the result of the query in string as JSON representation format. In the android script, the returned values are decoded to extract the values from it in JSON array to be used later by the android application.

V. CONCLUSIONS AND FUTURE WORK

Developing an effective patient appointment scheduling system is very important to ensure effective and efficient service delivery in medical centers and hospitals. The proposed mobile application for patient appointment scheduling is implemented to improve the services provided in the medical centers. Patients can register and book appointment in specific time without the need to visit the medical center, the system notify the patient before the booking time. Doctors can see the patient request and sends the notification to the patients if the appointment is available.

He can also view the comments given by the patient. The patient can login to his profile and view the medical history and the upcoming vaccinations and other information related to his health record.

The whole system was tested to certify its conformity with the design specification and requirement. The testing results show that the system is efficient, effective and interface is very user friendly. The application can handle acceptable number of requests at the same time. The results also show that the application provides very meaningful, accurate and timely information and provides privacy safety and for medical data of the patients.

This application may need further development to be implemented using multithreading to boost the application performance and to keep the application more responsive.

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